



NAF's Uzhavu Hackathon Shaastra 2025

Indian Institute of Technology (IIT), Madras

Solution Report

Livestock Health Management Platform

Team: AgroNext

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Submitted on: December 16, 2024

Abstract

Livestock management is a cornerstone of modern agriculture, vital for ensuring food security, economic stability, and environmental sustainability. However, traditional methods of monitoring animal health and managing breeding cycles are often labor-intensive, inefficient, and prone to errors. To address these challenges, this project proposes the development of a Livestock Health Management Platform, a comprehensive digital system designed to revolutionize livestock care through technology-driven solutions. The platform focuses on three key areas:

Health Monitoring: IoT devices enable real-time tracking of critical health parameters, facilitating early illness detection, reducing veterinary costs, and ensuring animal welfare.

Insemination Schedule Management: The system uses IoT sensors and machine learning models to monitor fertility windows, optimizing breeding cycles and conception rates.

Personalized Care Recommendations: By analyzing health and productivity data, machine learning algorithms provide tailored insights, including nutrition, vaccination, and treatment plans.

The solution employs a modular architecture, combining wearable sensors, environmental IoT devices, and a cloud-based analytics platform. A user-friendly interface ensures accessibility for farmers, with features like data visualization, automated alerts, and remote management. Predictive analytics further assist farmers in addressing health and breeding challenges proactively.

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1 Introduction

The management of livestock health and reproduction is a cornerstone of sustainable agriculture. With rising demands for high-quality animal products and increasing farm operational challenges, there is a growing need for advanced solutions to address inefficiencies in traditional livestock management practices. Farmers often grapple with issues such as monitoring individual animal health, tracking reproductive schedules, and catering to the diverse needs of livestock species. These challenges, if not addressed effectively, can lead to decreased productivity, elevated costs, and compromised animal welfare.

Traditional methods for monitoring livestock health rely heavily on manual labor and observational techniques, which are prone to inaccuracies and inefficiencies. Identifying early signs of illness, monitoring environmental stressors, and ensuring optimal feeding and care for each animal is often difficult in large-scale operations. Similarly, the lack of precise reproductive cycle tracking results in missed opportunities for artificial insemination or natural breeding, impacting overall herd productivity. In a competitive market, these limitations can significantly affect a farm's profitability and sustainability.

This report introduces a comprehensive solution—a Livestock Health Management Platform that integrates advanced digital technologies to transform livestock care. By leveraging the Internet of Things (IoT), machine learning, and cloud computing, the platform enables real-time health monitoring, efficient reproductive management, and personalized care recommendations tailored to individual animals. The proposed system addresses the complexity of livestock management by providing farmers with actionable insights, predictive analytics, and automated alerts, reducing manual intervention and enhancing decision-making.

Unlike generic solutions, this platform is designed with versatility, making it applicable to a range of livestock species common in Indian agriculture, including cattle, buffaloes, goats, and sheep. The system's modular approach allows farmers to adopt features based on their specific needs, ensuring both scalability and cost-effectiveness. For example, wearable devices monitor health parameters such as body temperature and heart rate, while reproductive tracking tools help optimize insemination schedules and fertility management.

By focusing on proactive and data-driven approaches, the Livestock Health Management Platform aims to enhance farm productivity, improve animal welfare, and promote sustainability in agriculture. This report details the design, implementation, and benefits of the system, providing a blueprint for integrating modern technology into traditional farming practices.

2 Problem Statement

The livestock industry faces a pressing challenge in modernizing its management practices to meet the growing demands for efficiency, productivity, and animal welfare. Despite advancements in technology, many farms still rely on manual and fragmented systems to monitor health, reproduction, and care, leading to inefficiencies and missed opportunities for optimization.

Key issues include the inability to systematically track individual animals' health and reproductive cycles, leading to delays in detecting diseases and suboptimal breeding practices. Farmers often lack access to timely and actionable insights, which are critical for making informed decisions about insemination timing, preventive care, and nutrition. This lack of precision contributes to economic losses and limits the potential for scaling operations sustainably.

Additionally, existing solutions are either too generalized to address the specific needs of different species or too costly and complex for small to medium-scale farms. As a result, there is a significant gap in accessible, user-friendly systems that can integrate diverse data sources, such as wearable IoT devices, environmental sensors, and farm records, into actionable insights tailored to individual farms.

Without addressing these challenges, the livestock sector risks falling short of its potential to meet the increasing demand for high-quality animal products while ensuring ethical and sustainable practices. The need for an innovative, scalable, and cost-effective solution has never been more urgent, making this project's focus on digital transformation both timely and essential.



Figure 1: A farm facing challenges in modern livestock management.

3 Proposed Solution

3.1 Overview

The proposed Livestock Health Management Platform is designed as a comprehensive digital ecosystem that addresses key challenges in livestock management. It combines real-time health monitoring, efficient insemination schedule management, and tailored care plans through a robust technological framework. The solution includes the following components:

- **IoT-Enabled Wearable Sensors:** Devices for collecting health-related data, such as heart rate and body temperature.
- **Centralized Data Management:** A secure database for storing, organizing, and retrieving livestock-related records.
- **Machine Learning Algorithms:** Tools for analyzing data patterns and generating actionable insights for optimal livestock care.
- **Mobile Application Interface:** A user-friendly platform to provide farm owners with real-time insights and management tools.

3.2 System Architecture

The platform's architecture is modular and scalable, ensuring seamless integration of IoT devices and software components. It is divided into three core subsystems:

- **Data Acquisition:** IoT sensors mounted on livestock capture critical health parameters and environmental factors.
- **Data Processing and Analysis:** A cloud-based server processes incoming data, identifies irregularities, and suggests interventions using machine learning.
- **User Interaction Layer:** An intuitive mobile application displays processed information, allowing farmers to monitor health and productivity trends effortlessly.

3.3 Insemination Schedule Management

To enhance reproductive efficiency, the platform uses advanced techniques to manage and optimize insemination schedules. This includes:

- **Estrous Cycle Monitoring:** Automated detection of heat cycles based on real-time behavioral and physiological parameters.
- **Ovulation Prediction:** Utilization of body temperature data and hormonal trends to identify precise ovulation periods.
- **Historical Analysis:** Leveraging past breeding data to refine predictions and improve conception success rates.

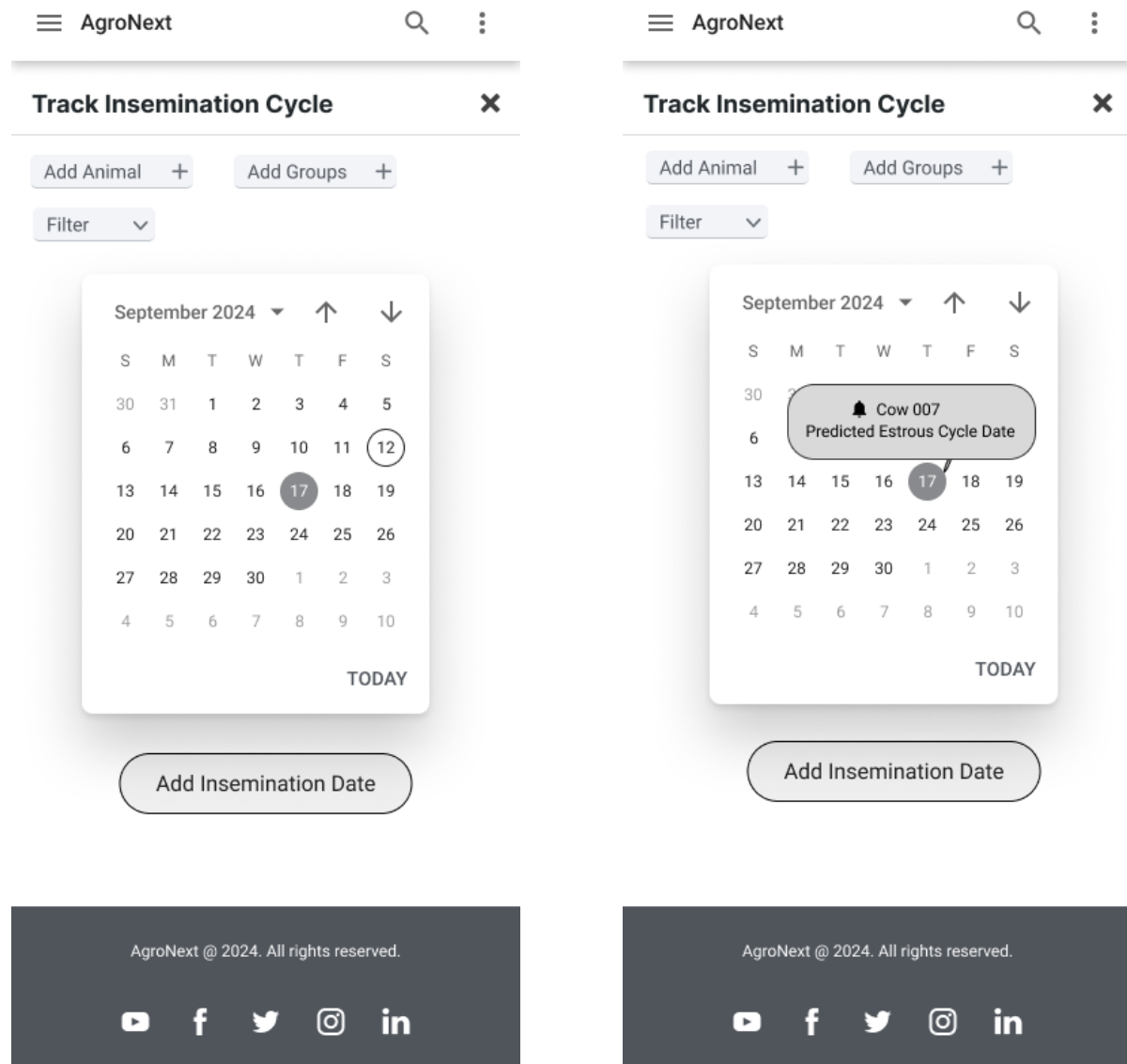


Figure 2: Insemination Schedule Management.

3.4 Personalized Care Plans

The platform delivers tailored care recommendations for each animal, ensuring comprehensive and efficient livestock management:

- Nutritional adjustments to improve productivity and health.
- Vaccination schedules based on disease susceptibility patterns.
- Alerts for timely interventions in case of illnesses or deviations in productivity metrics.

The screenshot displays the 'Add Insemination Date' form within the AgroNext mobile application. The form is titled 'Add Insemination Date' with a close button (X) in the top right corner. Below the title, there is a subtitle 'Cow Dashboard - Track your cow's insemination cycle'. The form contains several input fields: 'Name/Label' with a placeholder 'Enter Name', 'Breed' with a dropdown menu showing 'Breed of your animal', 'Sex' with a placeholder 'Male / Female', and 'Internal ID' with a placeholder 'Ex: Cow007'. Below these fields, there is a section titled 'Insemination' which includes a 'Date Of Insemination' field with a placeholder 'dd/mm/yyyy' and a calendar icon, and a 'Days from previous date of insemination' field with a placeholder 'Ex- 10 days'. At the bottom of the form, there is a dark grey footer containing the text 'AgroNext @ 2024. All rights reserved.' and social media icons for YouTube, Facebook, Twitter, Instagram, and LinkedIn.

Figure 3: Alerts and options for updating upcoming reproductive cycles.

3.5 Mobile Application Features

The mobile application acts as a command center for farm owners, enabling streamlined livestock management. Its key features include:

- **Real-Time Notifications:** Alerts for heat detection, illness symptoms, and critical insemination timings.

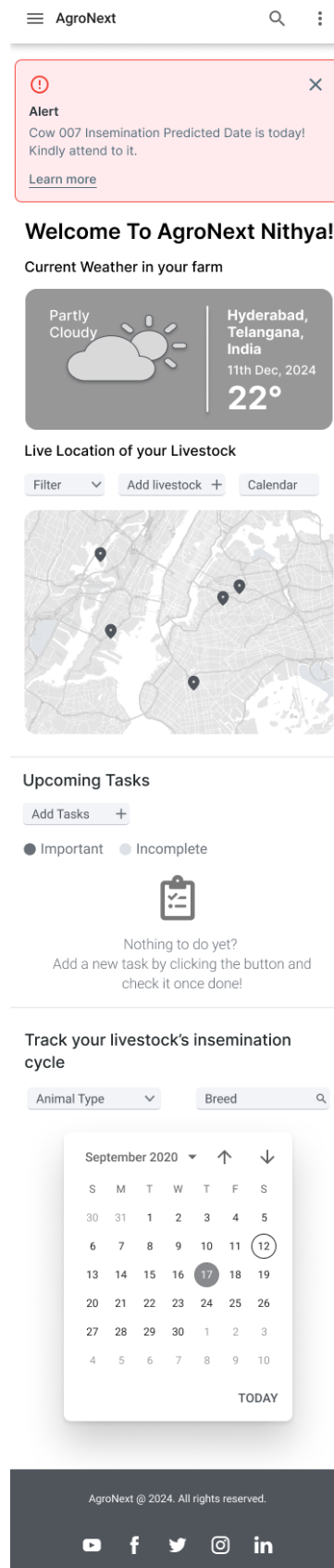
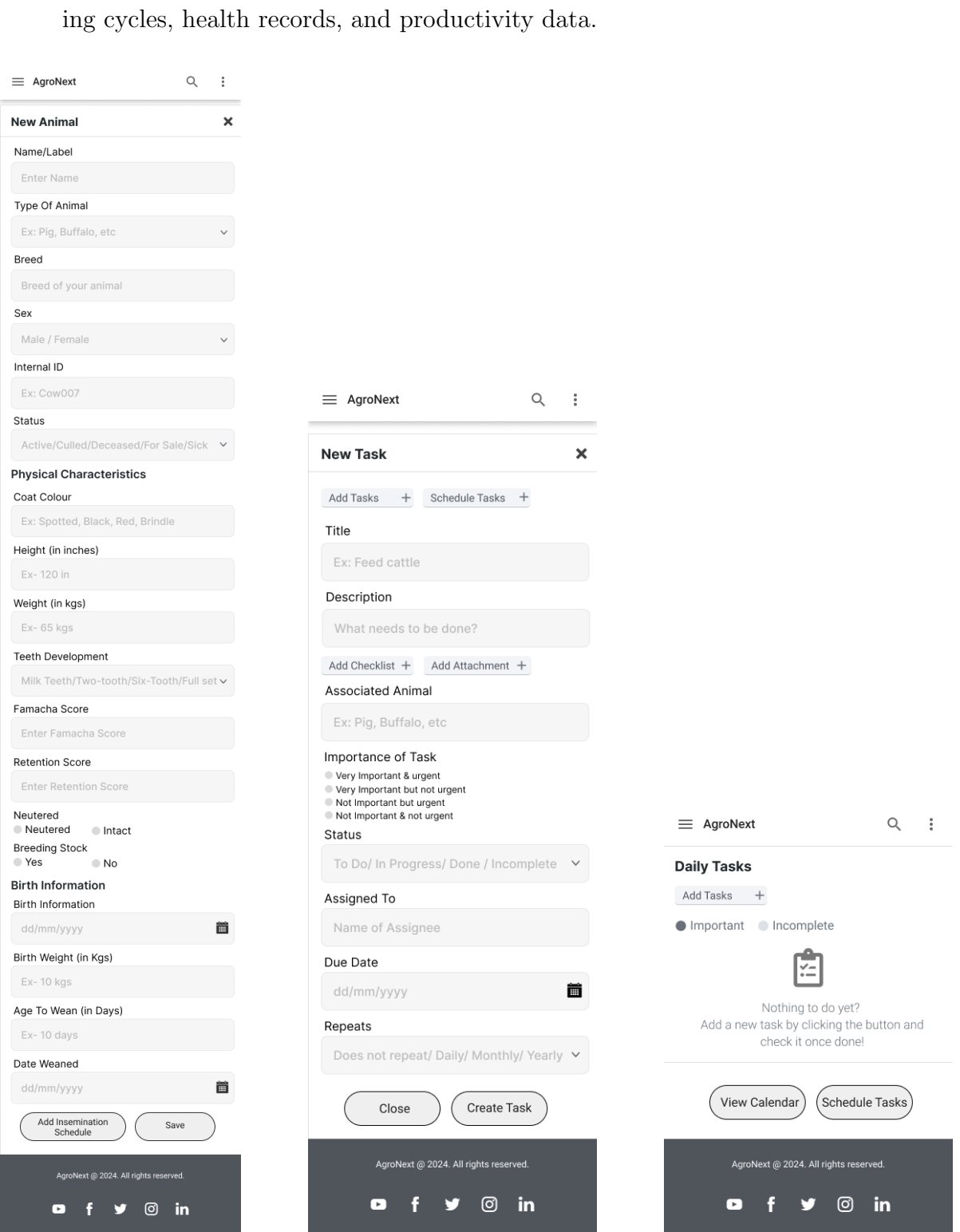


Figure 4: User Interface for Real-Time Notifications.

- **Comprehensive Data Logging:** Detailed animal profiles and logs of past breed-



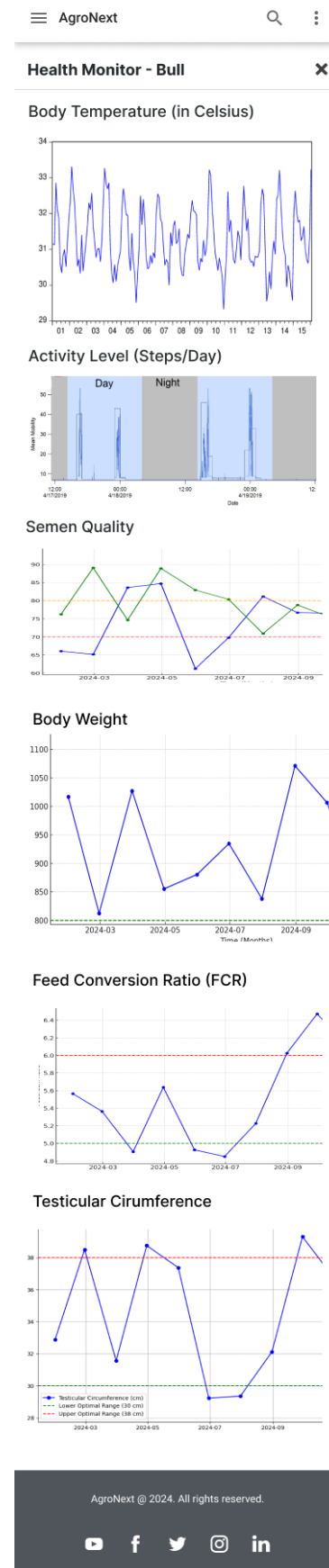
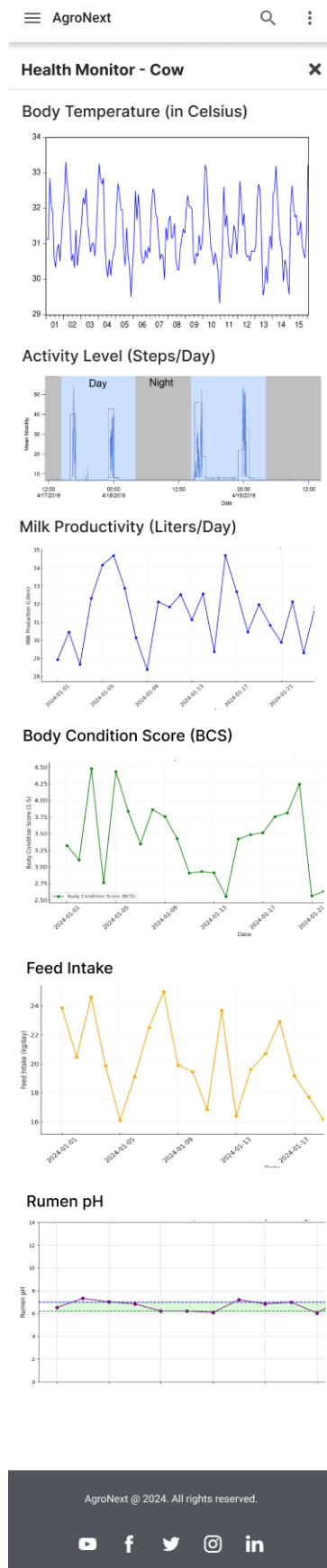


Figure 6: Predictive analytics dashboard for health and welfare insights.

4 Key Monitoring Parameters and IoT Integration

The integration of IoT technologies enables the continuous and automated monitoring of critical physiological, behavioral, and environmental parameters in livestock. This approach enhances the accuracy of data collection, facilitates real-time decision-making, and ensures more efficient management. The following outlines the key monitoring parameters and corresponding IoT integration strategies.

4.1 Estrous Cycle Monitoring

To ensure timely and efficient breeding cycles, IoT-enabled devices track and analyze key behavioral and physical indicators:

- **Behavioral Changes:** Activity tracking sensors, such as accelerometers and pressure-sensitive devices, detect mounting behavior, which serves as a reliable indicator of estrous activity.
- **Physical Changes:** Wearable sensors, including temperature and motion sensors, monitor physical changes such as temperature fluctuations and restlessness, which are associated with estrous periods.

4.2 Ovulation Timing

Accurately predicting ovulation timing is critical for optimizing breeding cycles. IoT technologies facilitate the precise tracking of the following:

- **Body Temperature:** IoT-based temperature sensors continuously monitor slight variations in body temperature, which correspond to the timing of ovulation, providing early warning signals for optimal insemination.
- **Hormonal Levels:** Advanced IoT devices, such as sensors integrated into blood or milk sampling systems, offer non-invasive monitoring of hormonal changes, further enhancing the precision of breeding schedules.

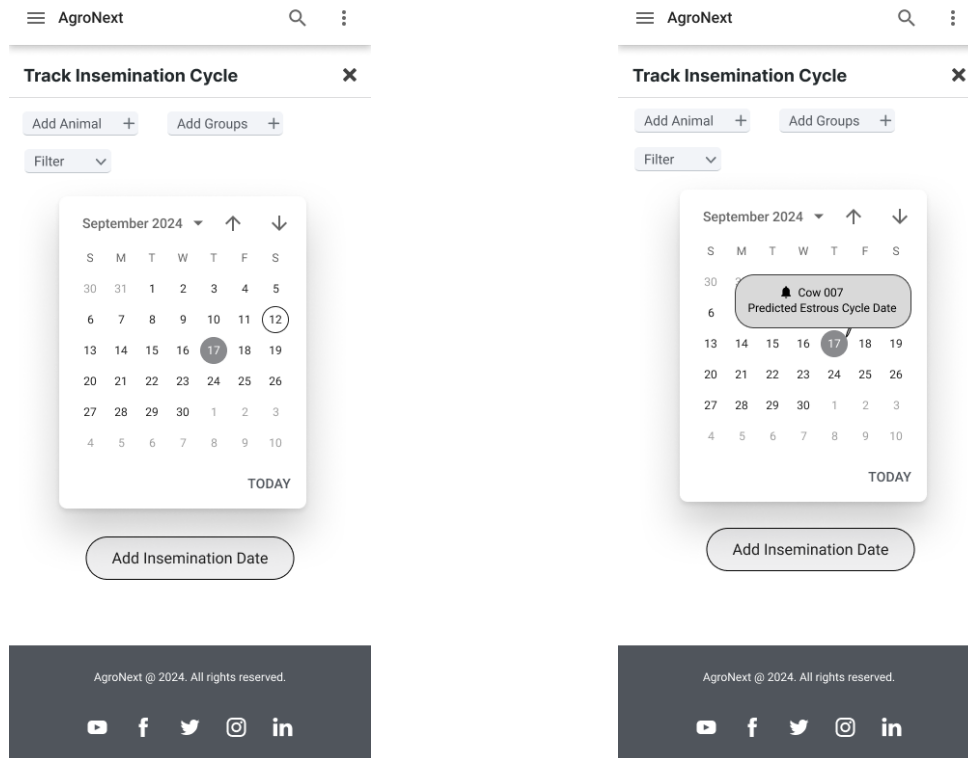


Figure 7: IoT tools for fertility monitoring and alerts for upcoming cycles.

4.3 Health Parameters

A healthy livestock population is paramount to farm productivity. IoT sensors are employed to monitor critical health parameters:

- **Body Condition Score (BCS):** High-resolution smart cameras equipped with image processing capabilities enable non-invasive body condition scoring to assess overall animal health and predict production potential.
- **Stress Indicators:** Sensors for heart rate and cortisol levels provide valuable insights into the animal's stress levels, which can affect reproductive performance, productivity, and overall well-being.

AgroNext

!

Alert
Cow 007 Insemination Predicted Date is today!
Kindly attend to it.
[Learn more](#)

Welcome To AgroNext Nithya!
Current Weather in your farm

Partly Cloudy



Hyderabad,
Telangana,
India
11th Dec, 2024
22°

Live Location of your Livestock
Filter Add livestock + Calendar



Upcoming Tasks
Add Tasks +
● Important ● Incomplete

Nothing to do yet?
Add a new task by clicking the button and check it once done!

Track your livestock's insemination cycle
Animal Type Breed

September 2020

↑ ↓

S	M	T	W	T	F	S
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	1	2	3
4	5	6	7	8	9	10

TODAY

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Figure 8: IoT integration for environmental monitoring and animal welfare tracking.

4.4 Environmental Factors

Managing environmental stressors, especially in intensive farming operations, is crucial to maintaining animal health and optimizing production. IoT-based solutions for environmental monitoring include:

- **Heat Stress:** Wearable heat sensors integrated with weather monitoring systems track environmental temperatures and humidity, allowing real-time detection of heat stress conditions, which significantly affect livestock health and productivity.
- **Seasonality:** IoT-controlled lighting systems, capable of simulating controlled photoperiods, optimize animal activity and reproductive cycles by regulating exposure to artificial light, effectively counteracting the seasonal impacts of daylight fluctuations.

The screenshot displays the AgroNext web application interface. On the left, a 'Welcome To AgroNext!' message is followed by a logo and a prompt: 'Lets get started! Give us details about your farm.' Below this is a form for entering farm details: 'Your Farm Name' (with a text input field), 'Where is your Farm Located?' (with sub-fields for 'Country', 'City', 'Address', 'State', and 'Pincode'), and two buttons: 'Add Livestock' and 'Next'. On the right, the 'Daily Tasks' section shows a header with 'AgroNext', a search icon, and a menu icon. Below the header is a 'Daily Tasks' section with an 'Add Tasks' button and a toggle for 'Important' (selected) and 'Incomplete'. A clipboard icon is shown with the text 'Nothing to do yet? Add a new task by clicking the button and check it once done!'. At the bottom of this section are 'View Calendar' and 'Schedule Tasks' buttons. The footer contains the text 'AgroNext @ 2024. All rights reserved.' and social media icons for YouTube, Facebook, Twitter, Instagram, and LinkedIn.

Figure 9: Farm location tracking and task scheduling for environmental management.

4.5 Additional IoT Features

Beyond the core health and environmental parameters, IoT technologies provide added functionalities that enhance farm management:

- **Genetic and Fertility Monitoring:** Advanced IoT tools assist in sire selection and optimize reproductive planning, helping farmers manage inbreeding and ensure genetic diversity in the livestock population.

- **Disease Detection:** Early-warning systems enabled by IoT devices help detect diseases such as mastitis or infections, ensuring swift intervention and minimizing the impact on herd health and farm productivity.

The image displays two screenshots of the AgroNext Livestock Management Platform interface.

Left Screenshot (Main Dashboard):

- Header: AgroNext logo, search icon, and menu icon.
- Section: Livestock Management
- Buttons: Add Animal (+), Add Group (+), Filter (v).
- Icon: A clipboard icon with a checkmark.
- Text: "No animals yet? Add a new animal by clicking the button and they'll show up here!"
- Buttons: View Calendar, Manage Animal.
- Footer: AgroNext @ 2024. All rights reserved. Social media icons (YouTube, Facebook, Twitter, Instagram, LinkedIn).

Right Screenshot (New Animal Form):

- Header: AgroNext logo, search icon, and menu icon.
- Section: New Animal (with close button X)
- Fields: Name/Label (Enter Name), Type Of Animal (Ex: Pig, Buffalo, etc), Breed (Breed of your animal), Sex (Male / Female), Internal ID (Ex: Cow007), Status (Active/Culled/Deceased/For Sale/Sick), Physical Characteristics (Coat Colour: Ex: Spotted, Black, Red, Brindle; Height (in inches): Ex- 120 in; Weight (in kgs): Ex- 65 kgs; Teeth Development: Milk Teeth/Two-tooth/Six-Tooth/Full set), Famacha Score (Enter Famacha Score), Retention Score (Enter Retention Score), Neutered (Neutered, Intact), Breeding Stock (Yes, No), Birth Information (Birth Information: dd/mm/yyyy, Birth Weight (in Kgs): Ex- 10 kgs, Age To Wean (in Days): Ex- 10 days, Date Weaned: dd/mm/yyyy).
- Buttons: Add Insemination Schedule, Save.
- Footer: AgroNext @ 2024. All rights reserved. Social media icons (YouTube, Facebook, Twitter, Instagram, LinkedIn).

Figure 10: IoT-enabled livestock data management and disease detection features.

5 Implementation Plan

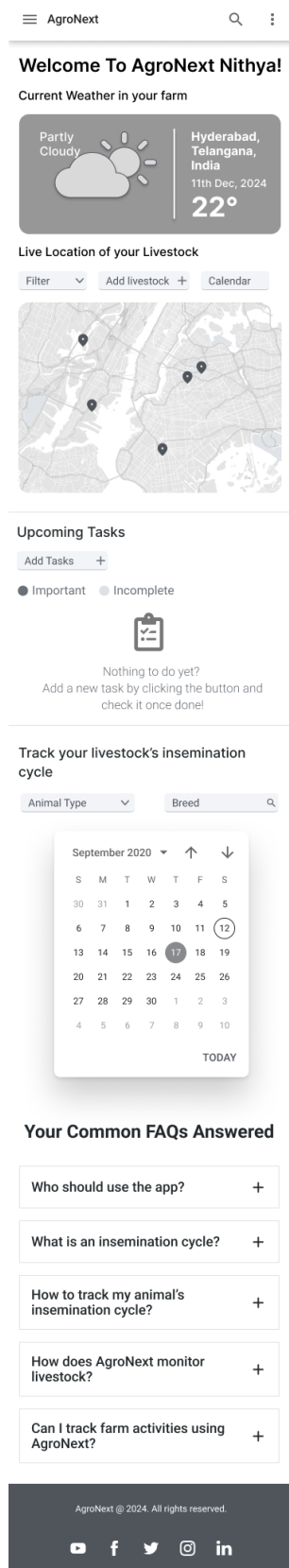
The implementation of the Livestock Health Management Platform is a structured process, designed to ensure the successful integration of both hardware and software components. The following sections outline the key technological components involved, as well as the phases of development that will guide the execution of the project.

5.1 Technological Components

The system's architecture is comprised of multiple critical components, each of which plays a pivotal role in enabling the platform's functionality. The key components are as follows:

- **Hardware:** The system utilizes IoT-enabled sensors and connectivity modules that are deployed on the livestock and environmental surroundings. These devices capture vital health and behavioral data, allowing real-time monitoring of the livestock.
- **Software:** The platform includes backend APIs that enable seamless communication between sensors, data storage, and user interfaces. A robust data analytics pipeline processes the raw sensor data to generate actionable insights, while a user-friendly mobile application interface presents these insights to farm managers for immediate action.

The integration of these components ensures smooth operation from data acquisition to decision-making.



AgroNext



Choose Your Preferred Language

English

தமிழ்

తెలుగు

हिन्दी

বাংলা

मराठी

മലയാളം

कोंकणी

Figure 11: Mobile interface showcasing system overview and language selection.

5.2 Phases of Development

The development process will be executed in a series of carefully planned phases, ensuring both technical feasibility and system effectiveness. The phases are as follows:

1. **Prototype Development and Sensor Integration:** In this initial phase, the hardware components, including IoT sensors and connectivity modules, will be integrated into a prototype system. This phase will focus on ensuring the devices function correctly and transmit data reliably.
2. **Backend Architecture Setup and Database Design:** The second phase will involve setting up the backend infrastructure to manage data storage and processing. This includes the establishment of a relational database and the design of APIs for efficient data retrieval and management.
3. **Implementation of Analytics Algorithms:** Machine learning algorithms and statistical models will be integrated into the backend. These algorithms will analyze health and behavioral data to provide actionable insights and predictive analytics.

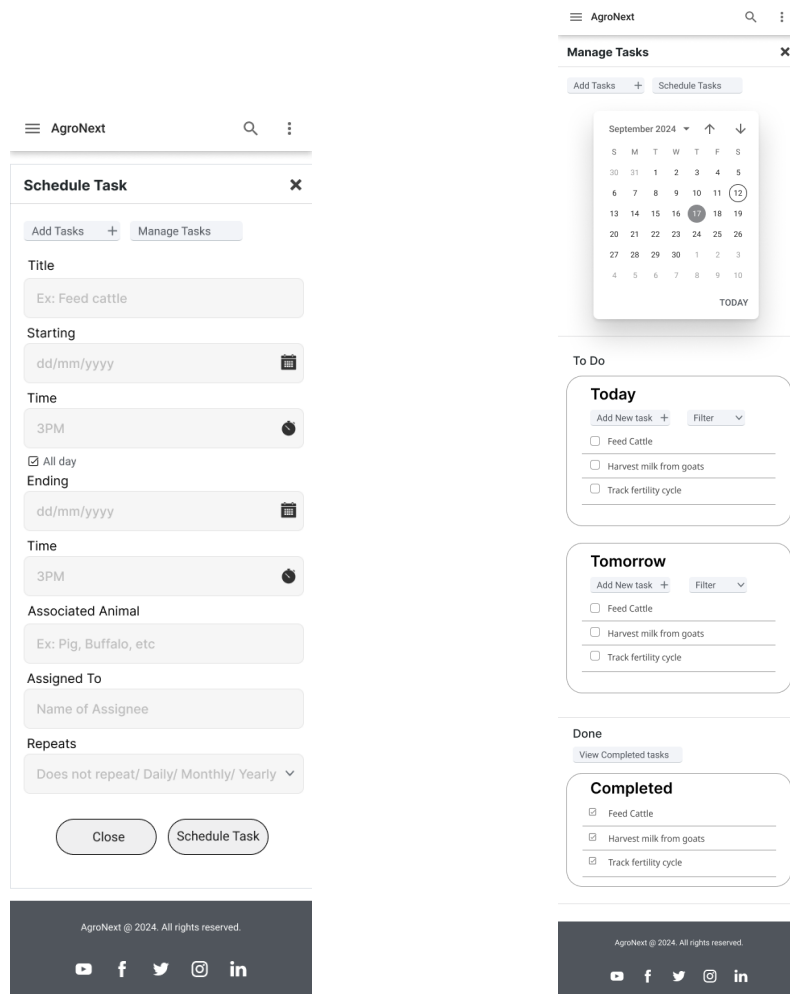


Figure 12: Integrating algorithms for scheduling tasks and task management.

4. **Development of Mobile Application:** The mobile application, acting as the primary interface for farmers, will be developed to allow users to view sensor data, receive real-time alerts, and manage animal care tasks seamlessly.

The 'New Animal' form in the AgroNext mobile application includes the following fields and options:

- Name/Label:** Text input field with placeholder 'Enter Name'.
- Type Of Animal:** Dropdown menu with placeholder 'Ex: Pig, Buffalo, etc'.
- Breed:** Text input field with placeholder 'Breed of your animal'.
- Sex:** Dropdown menu with placeholder 'Male / Female'.
- Internal ID:** Text input field with placeholder 'Ex: Cow007'.
- Status:** Dropdown menu with placeholder 'Active/Culled/Deceased/For Sale/Sick'.
- Physical Characteristics:**
 - Coat Colour:** Text input field with placeholder 'Ex: Spotted, Black, Red, Brindle'.
 - Height (in inches):** Text input field with placeholder 'Ex- 120 in'.
 - Weight (in kgs):** Text input field with placeholder 'Ex- 65 kgs'.
 - Teeth Development:** Dropdown menu with placeholder 'Milk Teeth/Two-tooth/Six-Tooth/Full set'.
 - Famacha Score:** Text input field with placeholder 'Enter Famacha Score'.
 - Retention Score:** Text input field with placeholder 'Enter Retention Score'.
 - Neutered:** Radio buttons for 'Neutered' and 'Intact'.
 - Breeding Stock:** Radio buttons for 'Yes' and 'No'.
- Birth Information:**
 - Birth Information:** Text input field with placeholder 'dd/mm/yyyy' and a calendar icon.
 - Birth Weight (in Kgs):** Text input field with placeholder 'Ex- 10 kgs'.
 - Age To Wean (in Days):** Text input field with placeholder 'Ex- 10 days'.
 - Date Weaned:** Text input field with placeholder 'dd/mm/yyyy' and a calendar icon.
- Buttons:** 'Add Insemination Schedule' and 'Save'.

Footer: AgroNext @ 2024. All rights reserved. Social media icons for YouTube, Facebook, Twitter, Instagram, and LinkedIn.

The 'New Task' form in the AgroNext mobile application includes the following fields and options:

- Buttons:** 'Add Tasks' and 'Schedule Tasks' with plus icons.
- Title:** Text input field with placeholder 'Ex: Feed cattle'.
- Description:** Text input field with placeholder 'What needs to be done?'.
- Buttons:** 'Add Checklist' and 'Add Attachment' with plus icons.
- Associated Animal:** Text input field with placeholder 'Ex: Pig, Buffalo, etc'.
- Importance of Task:** Radio buttons for 'Very Important & urgent', 'Very Important but not urgent', 'Not Important but urgent', and 'Not Important & not urgent'.
- Status:** Dropdown menu with placeholder 'To Do/ In Progress/ Done / Incomplete'.
- Assigned To:** Text input field with placeholder 'Name of Assignee'.
- Due Date:** Text input field with placeholder 'dd/mm/yyyy' and a calendar icon.
- Repeats:** Dropdown menu with placeholder 'Does not repeat/ Daily/ Monthly/ Yearly'.
- Buttons:** 'Close' and 'Create Task'.

Footer: AgroNext @ 2024. All rights reserved. Social media icons for YouTube, Facebook, Twitter, Instagram, and LinkedIn.

Figure 13: Mobile application features for adding animal details and tasks.

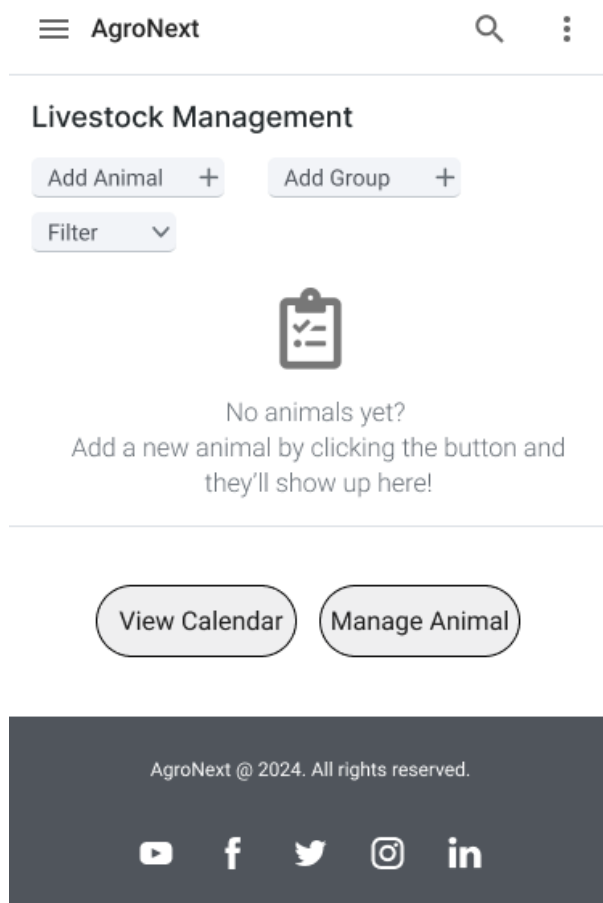


Figure 14: Livestock management functionality for data filtering and viewing.

5. **System Testing and Deployment:** The final phase includes comprehensive testing to identify potential issues and ensure the system is scalable, reliable, and secure. Once testing is completed, the system will be deployed for use in real-world farm environments.

5.3 Additional Features of the Platform

The mobile application will also include several general features and navigation tools for enhanced usability.

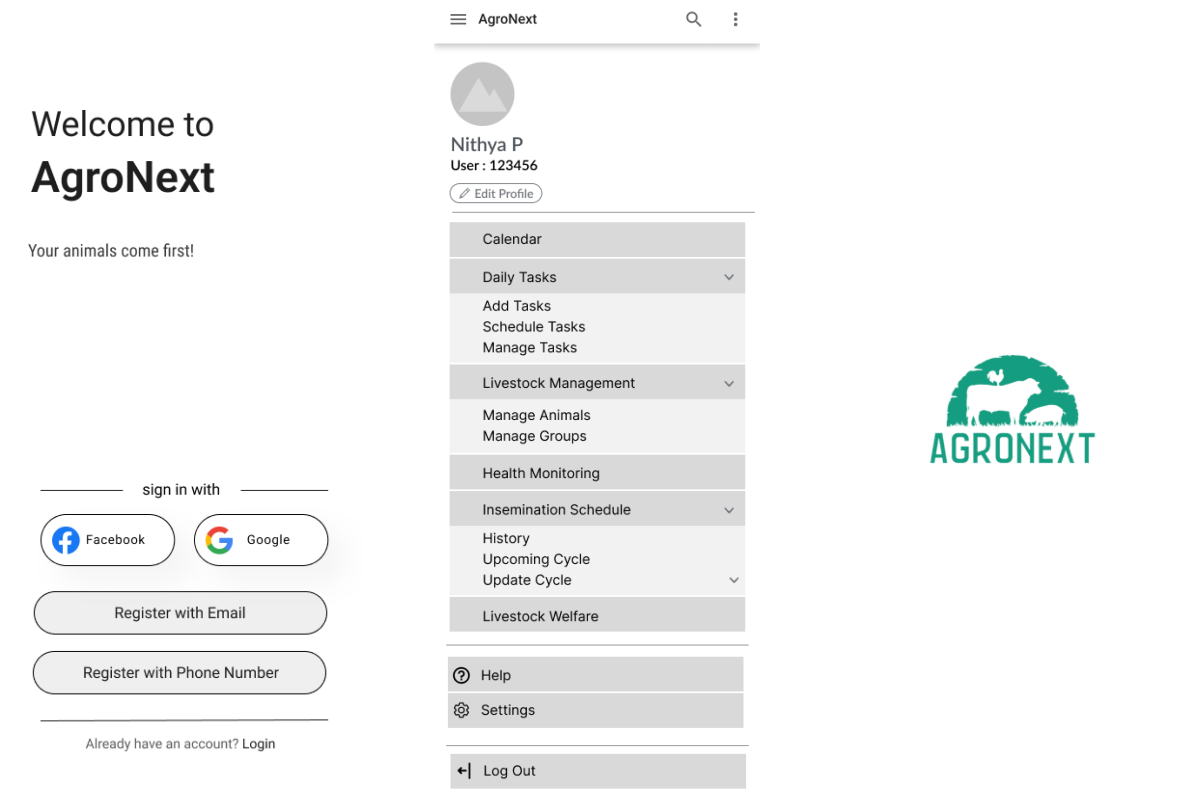


Figure 15: Overview of the application interface: Intro Screen, Menu Bar Options, and Splash Screen.

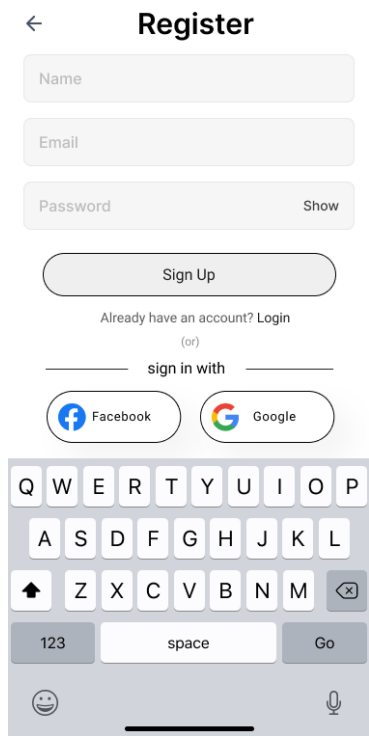


Figure 16: User registration process using email.

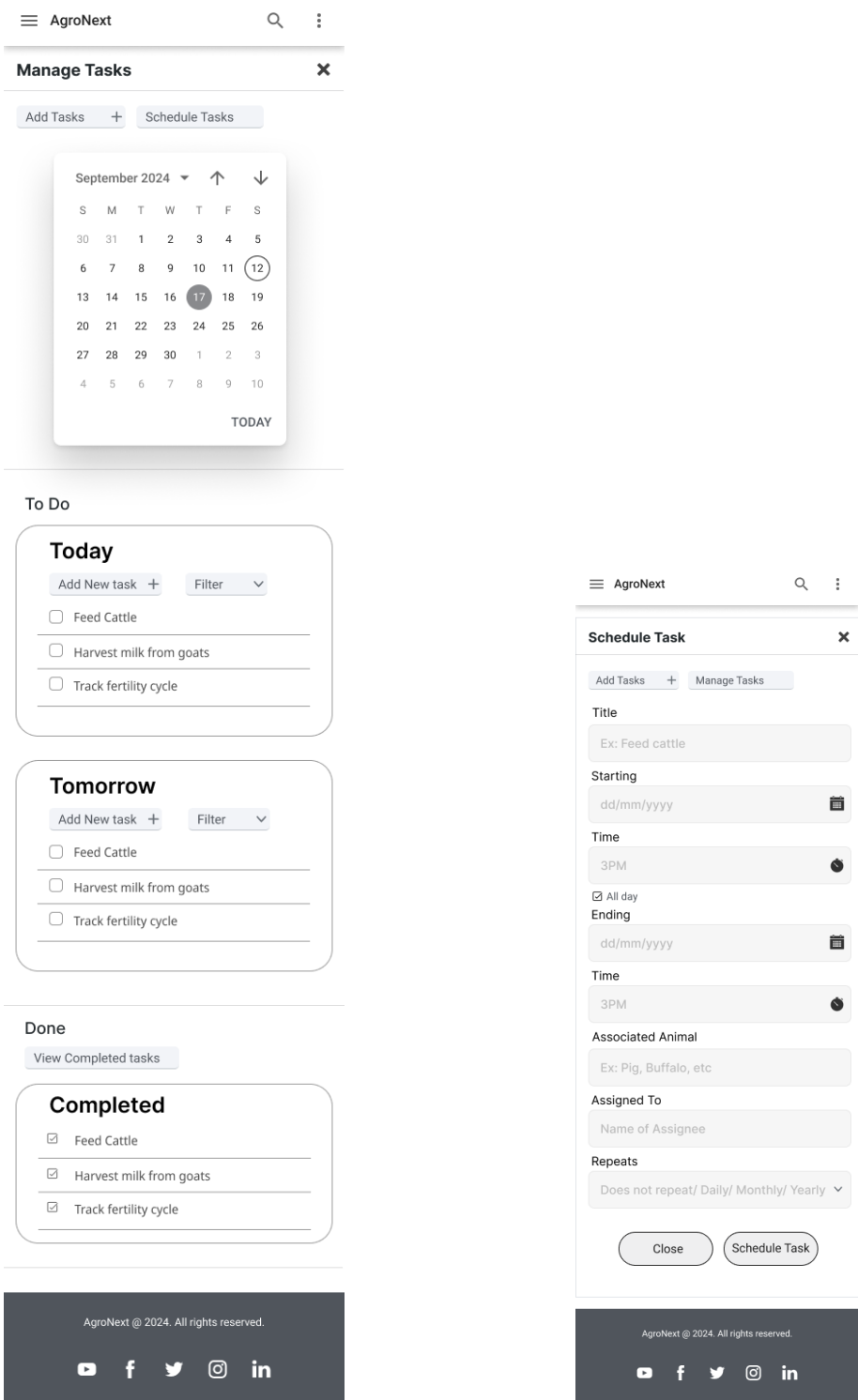


Figure 17: Task management and scheduling features.

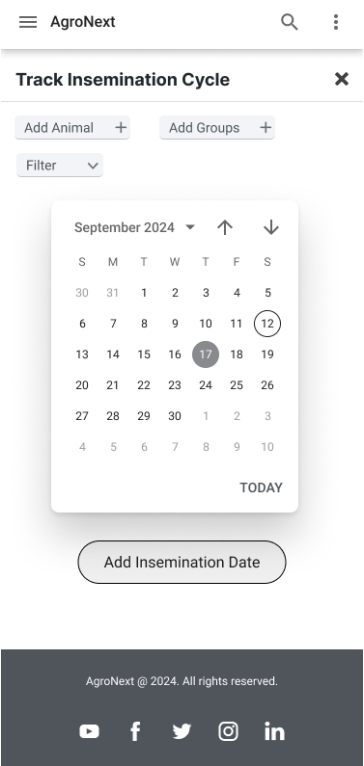


Figure 18: Insemination schedule displayed in a calendar format.

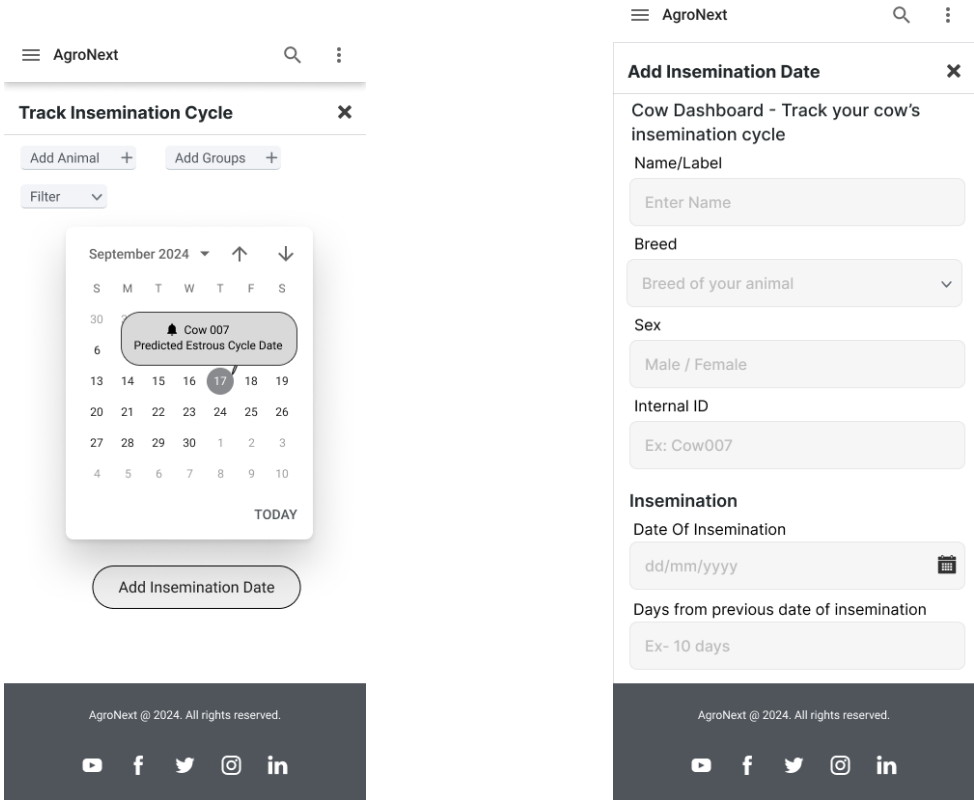


Figure 19: Alerts and options for updating upcoming reproductive cycles.

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⚠️

Alert
Cow 007 Insemination Predicted Date is today!
Kindly attend to it.
[Learn more](#)

✕

Welcome To AgroNext Nithya!
Current Weather in your farm

Partly Cloudy


Hyderabad,
Telangana,
India
11th Dec, 2024
22°

Live Location of your Livestock
Filter ▾ Add livestock + Calendar



Upcoming Tasks
Add Tasks +
● Important ● Incomplete

Nothing to do yet?
Add a new task by clicking the button and check it once done!

Track your livestock's insemination cycle
Animal Type ▾ Breed 🔍

September 2020 ▾ ↑ ↓

S	M	T	W	T	F	S
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	1	2	3
4	5	6	7	8	9	10

TODAY

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Figure 20: Animal welfare monitoring: Weather, location of livestock, etc.

6 Conclusion

The Livestock Health Management Platform represents a transformative approach to addressing the key challenges faced by the livestock farming industry. By leveraging advancements in IoT and data analytics, the platform integrates health monitoring, reproductive management, and personalized care into a unified, scalable solution.

The system's ability to provide real-time insights and actionable recommendations empowers farmers to make informed decisions, resulting in improved animal welfare, enhanced farm productivity, and reduced operational costs. Its modular and species-agnostic design ensures adaptability across diverse farming operations, from small-scale farms to large industrial setups.

This platform not only bridges the gap between traditional practices and modern technology but also sets the stage for sustainable and ethical livestock management. By addressing the critical aspects of animal health, reproduction, and care with precision and efficiency, it contributes to the broader goals of food security, economic resilience, and environmental sustainability. The Livestock Health Management Platform exemplifies the potential of digital innovation in revolutionizing agriculture, offering a robust foundation for the future of livestock management.

6.1 Supplementary Features

The image displays two side-by-side screenshots of the AgroNext web application. The left screenshot shows the 'Welcome To AgroNext!' login page. It features the AgroNext logo (a green silhouette of a cow and a sheep) and a form for entering farm details. The form includes fields for 'Your Farm Name', 'Country', 'City', 'Address', 'State', and 'Pincode'. Below the form are two buttons: 'Add Livestock' and 'Next'. The right screenshot shows the 'Daily Tasks' section of the application. It has a header with the AgroNext logo, a search icon, and a menu icon. Below the header is a 'Daily Tasks' section with an 'Add Tasks' button and a list of tasks. The tasks are categorized by 'Important' (red dot) and 'Incomplete' (blue dot). There is a clipboard icon and a message: 'Nothing to do yet? Add a new task by clicking the button and check it once done!'. At the bottom of the right screenshot are two buttons: 'View Calendar' and 'Schedule Tasks'. The footer of the right screenshot shows the copyright notice 'AgroNext @ 2024. All rights reserved.' and social media icons for YouTube, Facebook, Twitter, Instagram, and LinkedIn.

Figure 21: Farm location entry and daily task tracking features.